

0015 — BPS/STMR Exit-Rule Study: the exit signal IS the edge

Series: MC Setup Research Notes · **Session:** S73 (2026-07-14/15) · **Instrument:** SPX/SPXW options (signal on ES daily)

CONFIDENCE: HIGH — 142 trades over 14 years, identical pricing engine as the prior validated backtests (OptionsDX EOD bid/ask, real fees), and the result is a ranking of exit rules on the same entries, which is robust to most data issues.

TL;DR: The STMR bull-put-spread's profit comes **entirely from its timing exit** (first daily close > SMA5) — not from premium decay. Hold-to-expiry turns the strategy negative. The famous tastylive "manage winners at 50%" rule is ~flat here. Price stops are actively destructive. Separately, VIX-level/rank filters were tested and **rejected out-of-sample**: trade the strategy unconditioned. Practical rule: **enter on signal, exit on signal, no stops, no targets, never hold to expiry**.

1. The setup (self-contained)

- **STMR signal (entry):** ES daily stochastic %K8 < 15 AND Close > SMA100 — an oversold-pullback-in-uptrend mean-reversion signal (validated walk-forward as a futures system in note 0014: PF 4.45, 80% win, 16/17 years).
- **Options expression:** sell a SPX/SPXW **bull put spread** — short ~30Δ put, long put 50 pts lower, ~14 DTE. Defined risk = width – credit (~\$4–4.9K/lot).
- **Baseline exit:** first daily Close > SMA5 → buy the spread back; else cash settlement.
- **Pricing:** OptionsDX SPX EOD chains 2010–2023, **bid/ask fills** (sell at bid, buy at ask), \$1.30/contract/leg, settlement at intrinsic from official close.

2. Question

Does the exit rule matter — and specifically, would the industry-standard management rules (profit-take at 50% of credit; 2×credit stop; hold to expiration) beat the SMA5 signal exit? Secondly: does conditioning entries on VIX level/rank/direction improve results?

3. How we tested it

- Entries fixed: all 142 priceable daily-close STMR signals 2010–2023 (14 DTE / 50pt / 30Δ).
- The open spread was **re-priced at every EOD** it remained open (targeted per-strike scan of the chains — 12,312 strike-day marks over 2,588 sessions), giving a daily cost-to-close path per trade (buy short back at ask, sell long at bid).
- Six exit rules simulated on identical entries; early exits pay double fees.
- VIX study: each trade tagged with VIX close, 252-day rank, and 5-day change at entry (CBOE daily); in-sample filter table + tercile split + **yearly walk-forward** (train trailing 4y → pick best filter by expectancy (min 8 trades) → trade next year OOS).

4. Results

4.1 Exit-rule shootout (142 trades, bid/ask, \$1.30/ct, ES \$ values are per 1-lot)

Exit rule	Win %	PF	Avg \$/trade	Total	maxDD
SMA5 signal exit (baseline)	84%	1.74	+\$103	+\$14,668	-\$6,321
TP 50% of credit + SMA5	84%	1.72	+\$101	+\$14,298	-\$6,321
TP 50% of credit alone	87%	0.97	-\$12	-\$1,757	-\$18,056
Hold to expiry	82%	0.95	-\$29	-\$4,164	-\$27,740
TP50 + 2×credit stop	63%	0.71	-\$116	-\$16,495	-\$16,586
2×credit stop alone	58%	0.84	-\$77	-\$10,904	-\$18,239

4.2 VIX conditioning — in-sample filters (139 VIX-taggeable trades, full sample; descriptive only)

Filter at entry	n	Win %	PF	Avg	Total
none	139	84%	1.75	+\$105	+\$14,627
VIX rank > 33%	108	84%	1.44	+\$75	+\$8,143
VIX rank > 50%	82	83%	1.76	+\$106	+\$8,672
VIX falling (5d)	13	100%	∞	+\$335	+\$4,352
rank>33 & falling	8	100%	∞	+\$366	+\$2,931
VIX rank ≤ 33%	31	84%	6.15	+\$209	+\$6,484

By rank tercile: low 67.3%→PF 6.15 (n31) · mid PF 1.27 (n51) · high PF 1.67 (n57).

4.3 VIX walk-forward (train 4y → OOS next year)

Year	Filter chosen (in-sample)	OOS n	OOS win	OOS total
2014	rank>50	17	76%	+\$2,244

Year	Filter chosen (in-sample)	OOS n	OOS win	OOS total
2015	rank>50	7	71%	+\$1,413
2016	rank>50	3	100%	+\$937
2017	rank>33	8	100%	+\$3,163
2018	rank>33	1	0%	-\$3,355
2019	none	7	100%	+\$1,929
2020	none	9	78%	-\$599
2021	none	6	100%	+\$619
2022	none	6	83%	-\$61
2023	none	17	88%	+\$5,452

OOS filtered: n81, PF 2.13, +\$11,741 · OOS unfiltered (2014+): n87, PF 2.24, +\$12,948. The filter UNDERPERFORMS no-filter out-of-sample.

5. Why it works / fails

The STMR trade is a **mean-reversion timing play expressed in options**, not a theta play. The SMA5 exit captures the bounce and leaves before the next leg down; every management rule that overrides it degrades the trade in a diagnosable way:

- **TP50** exits winners before the bounce completes AND leaves losers unmanaged to expiry — it keeps the worst of both worlds.

- **Price stops** trigger during the drawdown that precedes the bounce being traded — they systematically sell the local bottom (win rate collapses 84%→58%).

- **Hold-to-expiry** re-exposes the position for ~2 weeks after the bounce is done; the tail events land in that window (maxDD quadruples to -\$27.7K).

- **VIX filters** mostly re-time the same trades and, per fold, latch onto whatever the

last 4 years rewarded; 2018's single-trade disaster shows the failure mode. The "elevated-but-falling IV" folk prior is unsupported (n≤13 buckets); if anything the low-VIX tercile is cleanest — consistent with mean-reversion working best in orderly regimes.

6. Recommendation

Trade the BPS exactly as specified: enter on the causal 15:59 STMR signal, exit on the first 15:59 spot > SMA5, otherwise settle. NO price stops, NO profit targets, NO expiry holds, NO VIX filter. The defined-risk structure (width – credit) is the only stop. Forward-test live on the paper pipeline (running since 2026-07-14).

7. Caveats

- Daily-close signal set (research view); the executable causal-15:59 set differs on ~13% of signals (see S70) — rankings should transfer, absolute totals may not.

- EOD marks understate intraday mark extremes; stop rules were evaluated on EOD closes, which if anything flatters stop rules (intraday stops would trigger more often).

- 2010–2023 window contains only two major vol events; expiry-hold tail risk is likely understated.

- Fees modeled at \$1.30/contract; live SPX all-in is similar but verify vs IB statements.

8. Reproduce

```
.venv/Scripts/python.exe scripts/mr_bps_exit_rules.py      # 4.1
.venv/Scripts/python.exe scripts/mr_bps_regime_wf.py       # 4.2 / 4.3
```

Data: data/optionsdx/.txt, data/ES_stoch_daily.csv, data/vix_daily.csv. Trade files: data/options_sim/bps_regime_trades.csv.